

FLIGO - Free Screen Recorder for PC

Open-Source Desktop Recording Studio with Dual Screen-Cam Overlay

PLATFORM: Windows 10/11

LICENSE: Free & Open-Source (GPL/MIT)

VERSION: v1.0

STATUS: Approved

1. PROJECT SCOPE & OBJECTIVE

Fligo is a 100% free, lightweight, open-source desktop screen recording application designed for creators, educators, and gamers. It features smart region selection, movable/resizable webcam overlays (Circle/Rect), and integrated real-time background noise suppression & auto-silence removal without watermarks or limits.

2. CORE FUNCTIONAL SPECIFICATIONS

FR-01: CUSTOMISABLE SCREEN REGION SELECTOR (WITH SOUND)

CORE CAPTURE

- Full Screen & Multi-Monitor Selection: Record entire display or switch between active monitors.
- Custom Drag-and-Drop Area Selector: Interactive on-screen bounding box to choose exact pixels.
- Application / Window Snap: Snap directly to active windows (e.g. Browser, IDE, Game).
- Preset Aspect Ratios: 16:9 (YouTube), 9:16 (Shorts/Reels), 1:1 (Square), and 4:3 presets.
- High-FPS & Resolution: 30 FPS / 60 FPS recording with 720p, 1080p Full HD, 2K, and 4K support.
- Synchronized System Sound: Native WASAPI loopback internal sound capture in lockstep with video.

FR-02: SOUND RECORDER (FREE NOISE & SILENCE REMOVAL)

AUDIO ENGINE

- Dual Audio Mixing: Record both PC System Audio and Microphone narration simultaneously.
- Free Real-Time AI/DSP Noise Suppression: Eliminates fan hum, AC hiss, keyboard clicks, and noise.
- Automatic Silence Removal: Detects pauses (< -40dB for > 1s) and automatically trims dead silence.
- Audio Gain Booster & VU Meter: Real-time visual audio level feedback with input volume boost.
- Synchronization Offset Tool: Calibration slider ensuring zero audio-video lag/desync.
- Formats Export: Standalone high-quality audio recording export to MP3, WAV, or AAC.

FR-03: WEBCAM RECORDER (STANDALONE MODE)

CAMERA MODE

- Dedicated Camera Mode: Standalone recording from USB webcams, virtual cams, and capture cards.
- High Resolution & FPS: Full HD 1080p and 4K recording up to 60 FPS.
- Live Camera Preview: Controls for horizontal flip/mirroring and aspect ratio adjustment.
- Bound Noise-Cancelled Audio: Integrates directly with the noise-suppressed microphone channel.

FR-04: DUAL MODE (SCREEN + MOVEABLE & RESIZABLE WEBCAM)

FLAGSHIP FEATURE

Simultaneous capture of screen recording and interactive floating webcam overlay with real-time controls:

Movable Overlay	Floating picture-in-picture camera box stays on top and can be dragged to ANY position on the screen in real-time during recording.
Shape Morphing	Instant switch between 3 custom geometric camera frames: Circle (Avatar style), Rounded Rectangle, and Classic 16:9 Rectangle.
Dynamic Resizing	Smooth real-time scale slider and corner drag handles (Small, Medium, Large, Custom % scaling) with optional glow/border styling.
AI Audio Engine	Combined screen sound and narrator voice processed with real-time noise suppression & automatic silence removal.

FLIGO - TECHNICAL SPECIFICATIONS & ARCHITECTURE

System Requirements, Non-Functional Metrics & Project Roadmap

3. UI CONTROLS & SHORTCUTS

Minimal Floating Control Bar	Clean on-screen widget with Start, Pause, Resume, Stop, Webcam toggle & Screenshot.
Hotkey: F9	Global shortcut to Start or Stop screen recording immediately.
Hotkey: F10	Global shortcut to Pause or Resume ongoing recording without opening UI.
Hotkey: F11	Instant toggle to Show or Hide floating webcam overlay.
System Tray Integration	Minimizes silently to system tray for distraction-free background recording.
Countdown Timer	Visual 3-second animated countdown before video capture begins.

4. NON-FUNCTIONAL & QUALITY ATTRIBUTES

Performance & GPU Accel	CPU utilization < 10% on 1080p 60fps recording via NVENC, Intel QuickSync, or AMD AMF hardware acceleration.
Zero Audio-Video Desync	Precision timestamp synchronization ensuring zero drift between mic, system audio, and video over 3+ hours.
100% Offline & Private	Zero telemetry, no tracking, no cloud upload requirement. All data remains strictly local on user's PC.
Free Noise Suppression	Embedded DSP/RNNoise spectral model requiring zero paid API keys or subscription fees.
Crash-Safe Containers	Option to record directly into MKV/MP4 formats so video remains recoverable even if PC shuts down.

5. SYSTEM ARCHITECTURE & COMPONENTS

UI & Control Layer	Capture & Compositor	Audio & DSP Engine
Tech: Electron / Tauri / PyQt - Floating toolbar widget - Region selection canvas - Global hotkey listeners - Tray menu controller	Tech: DirectX / WGC / OpenCV - Screen capture 60fps - Webcam feed masking - Circle/Rect crop shader - GPU hardware encoder	Tech: WASAPI & RNNoise - System sound loopback - Microphone audio input - Real-time noise removal - Auto silence jump-cutter

6. DEVELOPMENT ROADMAP & MILESTONES

Phase 1: Foundation	Implement Desktop Duplication capture & interactive drag-and-drop region selector.	Weeks 1-2
Phase 2: Sound & DSP	WASAPI loopback audio mixing + integrate free RNNoise & silence trimming.	Weeks 3-4
Phase 3: Dual Mode	Floating movable overlay, Circle & Rectangle shape masks, and resize slider.	Weeks 5-6
Phase 4: Packaging	Global hotkeys, settings panel, standalone .exe installer, and GitHub open-source release.	Weeks 7-8

SPECIFICATION APPROVAL & SUMMARY

This document constitutes the formal Software Requirements Specification for the Fligo Screen Recorder. All core requirements (custom screen selector, free noise removal, auto silence removal, camera recorder, and dual mode movable & shape-changeable webcam) have been fully specified for development.